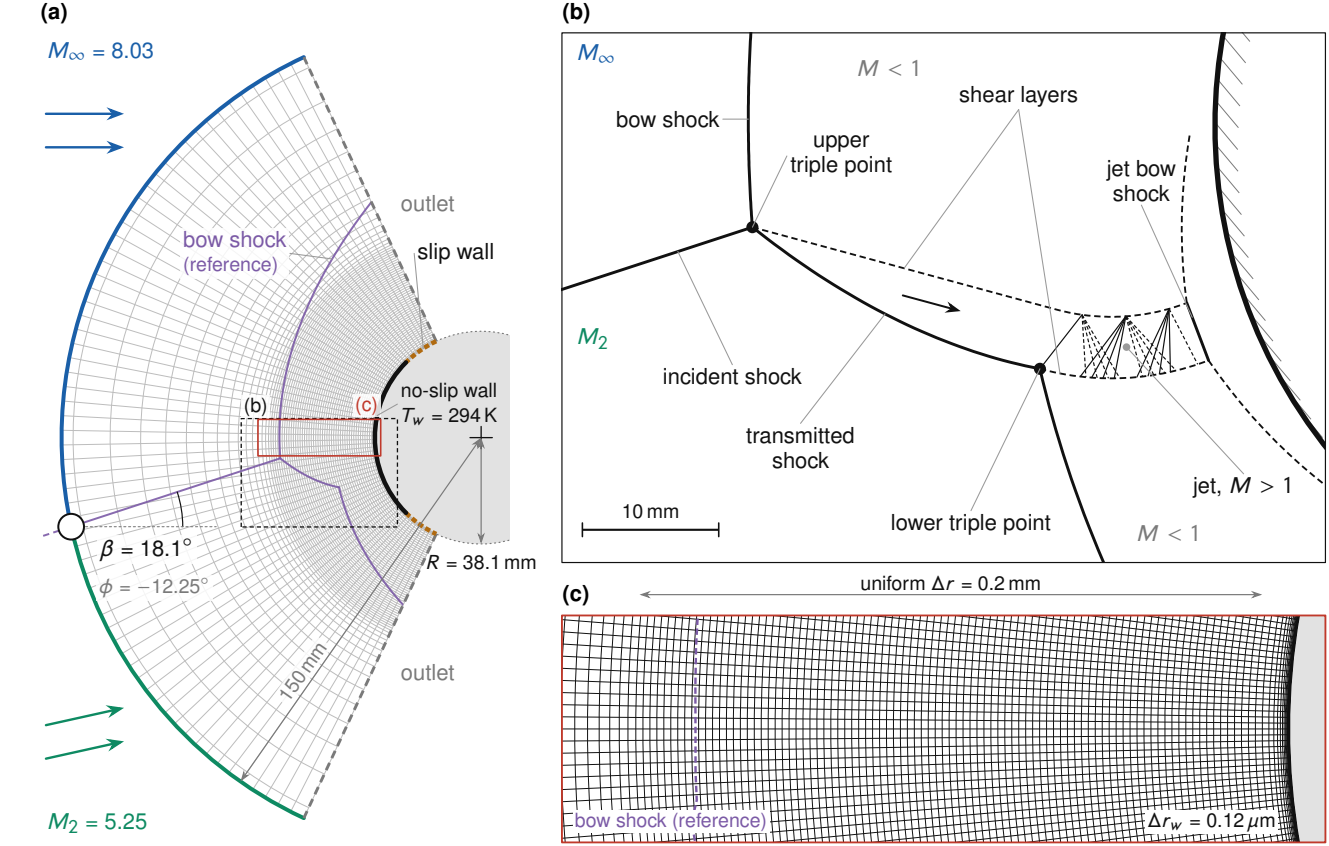

(a)
M_∞ = 8.03
outlet
bow shock
(reference)
slip wall
no-slip wall
T_w = 294 K
(b)
(c)
R = 38.1 mm
β = 18.1°
ϕ = −12.25°
150 mm
outlet
M_2 = 5.25
(b)
M_∞
M < 1
shear layers
bow shock
upper
triple point
jet bow
shock
M_2
incident shock
transmitted
shock
jet, M > 1
10 mm
lower triple point
M < 1
(c)
uniform Δr = 0.2 mm
bow shock (reference)
Δr_w = 0.12 μm